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To the Editors, Biomedical Signal Processing and Control

Dear Editors,

Please consider our manuscript "Mean correction in EEG decoding: decision geometry, temporal memory, and failure modes" for publication as a research article in Biomedical Signal Processing and Control.

The paper asks a question that every cross-session EEG pipeline answers implicitly and few check: when the mean of the target session is subtracted before decoding, does that help or harm? For a fixed affine decoder only the projection of the displacement onto the centred classifier weights can change a decision. In a symmetric two-class model the balanced-optimal threshold does not depend on the target class proportions, so a correction that removes a composition-only displacement must hurt, while a correction that removes a genuine drift can help; the optimal weight follows from the ratio of the two.

Three results carry the paper. First, the sign is a property of the benchmark: full-strength past-only correction gains 0.214 on SEED and 0.071 on BCI-2a and loses 0.102 on sleep-EDF, negative in all 32 subjects. Second, the harm is located rather than merely observed: the displacement the estimator actually subtracts is decomposed exactly, with the terms anchored on the source mean so that they do not depend on the feature origin, and the composition term carries the loss (-0.0715) while the within-class term is beneficial (+0.0202). A controlled factorial separates the two factors and places the measured optimum correction strength below one when composition is present. Third, the protocol matters as much as the estimator: a two-factor audit separates causal filtering from the alignment reference, and a negative case in grasp-and-lift did not survive an audit of its cache.

The work is methodological rather than architectural, which is why we believe it fits this journal: it concerns signal processing choices (filter causality, alignment reference, block memory, trial order) and the conditions under which they change a decoding result, with all code, per-subject outputs and negative results archived. Theory is used to explain the observed signs, not as a substitute for them, and the two independent corpora we include are reported with intervals rather than as confirmation.

The manuscript is original, is not under consideration elsewhere, and all authors have approved this submission. The authors declare no competing interests. The study uses public, de-identified datasets and required no new ethics approval. All figures are data plots produced by the archived scripts with Matplotlib; the layouts of Fig. 1(a) and Fig. 2(a) were drafted with AI assistance and re-rendered from the archived numbers, and no figure was produced by an image-generation model.

Yours sincerely,

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